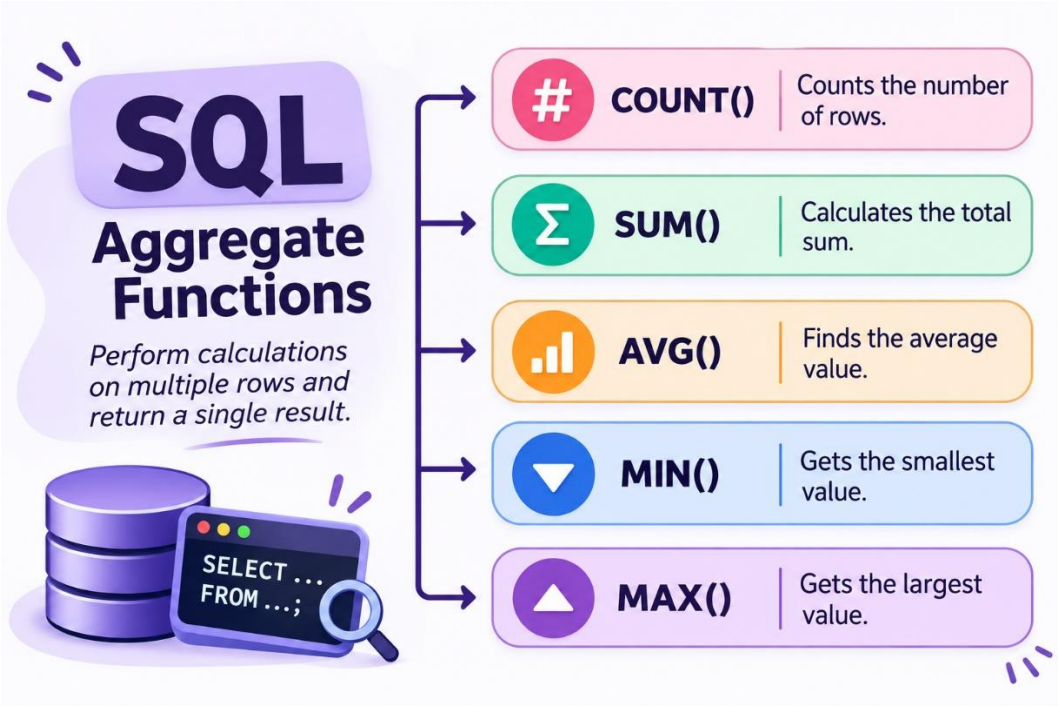
DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

WEEK 9

1. AGGREGATE FUNCTIONS

SQL Aggregate Functions are used to perform calculations on a group of rows and return a single result. Huft uses Aggregate Functions to analyses its online book sales by calculating the total orders, total sales revenue, average order value, and the highest and lowest order amounts.



1.1 COUNT() – Total Number of Orders

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM HuftOrder;

TOTAL_ORDERS
------------
          10
```

1.2 SUM() – Total Sales Amount

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM HuftOrder;

TOTAL_SALES
-----------
      12289
```

1.3 AVG() – Average order amount

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM HuftOrder;

AVERAGE_ORDER_AMOUNT
--------------------
              1228.9
```

1.4 MIN() – Minimum Order Amount

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order_Amount
  2  FROM HuftOrder;

MINIMUM_ORDER_AMOUNT
--------------------
                650
```

1.5 MAX() – Maximum Order Amount

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order_Amount
  2  FROM HuftOrder;

MAXIMUM_ORDER_AMOUNT
--------------------
                1849
```

2. GENERATE TOTAL SALES REPORT

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Sales
  5  FROM HuftOrder;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES
------------ ----------- -------------
          10       12289        1228.9
```

2.1 Customer-wise Sales Report

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase_Amount
  5  FROM CUST3 c
  6  JOIN HuftOrder o
  7      ON c.CID = o.Customer_ID
  8  GROUP BY c.CID, c.CNAME
  9  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID CUSTOMER_NAME                 TOTAL_PURCHASE_AMOUNT
----------- ----------------------------- ---------------------
          2 Anu                                            1849
          1 Abi                                            1750
          8 Meena                                          1498
          4 Karthik                                        1273
          7 Arun                                           1197
          3 Ravi                                           1099
          5 Priya                                          1075
         10 Nisha                                           900
          6 Divya                                           650

9 rows selected.
```

3. Top Customers Based on Purchase Amount

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM CUST3 c
  6  JOIN HuftOrder o
  7      ON c.CID = o.Customer_ID
  8  GROUP BY c.CID, c.CNAME
  9  ORDER BY Total_Purchase DESC
 10  FETCH FIRST 5 ROWS ONLY;

CUSTOMER_ID CUSTOMER_NAME                 TOTAL_PURCHASE
----------- ----------------------------- --------------
          2 Anu                                     1849
          1 Abi                                     1750
          8 Meena                                   1498
          4 Karthik                                 1273
          7 Arun                                    1197
```

4. Best-Selling Products

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      SUM(oi.Quantity) AS Total_Quantity_Sold
  5  FROM HuftProduct p
  6  JOIN HuftOrder_Item oi
  7      ON p.Product_ID = oi.Product_ID
  8  GROUP BY p.Product_ID, p.Product_Name
  9  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID PRODUCT_NAME              TOTAL QTY SOLD
---------- ------------------------- --------------
       107 Dental Chew Sticks                     4
       109 Puppy Training Pads                    3
       103 Leather Dog Collar                     3
       101 Chicken Dog Food                       3
       102 Grain Free Cat Food                    3
       108 Pet Feeding Bowl                       2
       112 Cat Grooming Brush                     2
       104 Pet Grooming Shampoo                   2
       106 Cat Litter                             2
       110 Pet Conditioner                        1

10 rows selected.
```

4.1 Product Sales – Amount and Quantity

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      SUM(oi.Quantity) AS Total_Quantity_Sold,
  5      SUM(oi.Total_Amount) AS Total_Sales_Amount
  6  FROM HuftProduct p
  7  JOIN HuftOrder_Item oi
  8      ON p.Product_ID = oi.Product_ID
  9  GROUP BY p.Product_ID, p.Product_Name
 10  ORDER BY Total_Sales_Amount DESC;

PRODUCT_ID PRODUCT_NAME              TOTAL QTY SOLD TOTAL_SALES_AMOUNT
---------- ------------------------- -------------- ------------------
       102 Grain Free Cat Food                    3               2247
       101 Chicken Dog Food                       3               1950
       107 Dental Chew Sticks                     4               1596
       109 Puppy Training Pads                    3               1497
       103 Leather Dog Collar                     3               1350
       106 Cat Litter                             2               1100
       104 Pet Grooming Shampoo                   2                700
       112 Cat Grooming Brush                     2                650
       108 Pet Feeding Bowl                       2                550
       110 Pet Conditioner                        1                425

10 rows selected.
```

5. Category-wise Sales Analysis

```
SQL> SELECT
  2      c.Category_ID,
  3      c.Category_Name,
  4      SUM(oi.Quantity) AS Total_Quantity_Sold,
  5      SUM(oi.Total_Amount) AS Total_Sales
  6  FROM HuftCategory c
  7  JOIN HuftProduct p
  8      ON c.Category_ID = p.Category_ID
  9  JOIN HuftOrder_Item oi
 10      ON p.Product_ID = oi.Product_ID
 11  GROUP BY c.Category_ID, c.Category_Name
 12  ORDER BY Total_Sales DESC;

CATEGORY_ID CATEGORY_NAME                                      TOTAL QTY SOLD TOTAL_SALES
----------- -------------------------------------------------- -------------- -----------
          1 Dog Food                                                       10        5043
          2 Cat Food                                                        5        3347
          3 Accessories                                                     5        1900
          4 Grooming                                                        5        1775
```

6. Combined Business Sales Report

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  6      p.Product_Name,
  7      cat.Category_Name,
  8      oi.Quantity,
  9      oi.Total_Amount AS Item_Amount,
 10      o.Order_Status
 11  FROM CUST3 c
 12  JOIN HuftOrder o
 13      ON c.CID = o.Customer_ID
 14  JOIN HuftOrder_Item oi
 15      ON o.Order_ID = oi.Order_ID
 16  JOIN HuftProduct p
 17      ON oi.Product_ID = p.Product_ID
 18  JOIN HuftCategory cat
 19      ON p.Category_ID = cat.Category_ID
 20  ORDER BY o.Order_ID;
rows will be truncated

rows will be truncated

rows will be truncated
```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DATE	PRODUCT_NAME	CATEGORY_NAME
1	Abi	1001	18-SEP-2026	Chicken Dog Food	Dog Food
1	Abi	1001	18-SEP-2026	Leather Dog Collar	Accessories
2	Anu	1002	18-SEP-2026	Grain Free Cat Food	Cat Food
2	Anu	1002	18-SEP-2026	Cat Litter	Cat Food
3	Ravi	1003	19-SEP-2026	Pet Grooming Shampoo	Grooming
3	Ravi	1003	19-SEP-2026	Dental Chew Sticks	Dog Food
4	Karthik	1004	19-SEP-2026	Puppy Training Pads	Dog Food
4	Karthik	1004	19-SEP-2026	Pet Feeding Bowl	Accessories
5	Priya	1005	20-SEP-2026	Cat Grooming Brush	Grooming
5	Priya	1005	20-SEP-2026	Pet Conditioner	Grooming
6	Divya	1006	20-SEP-2026	Chicken Dog Food	Dog Food
7	Arun	1007	20-SEP-2026	Dental Chew Sticks	Dog Food
8	Meena	1008	21-SEP-2026	Grain Free Cat Food	Cat Food
10	Nisha	1010	21-SEP-2026	Leather Dog Collar	Accessories

14 rows selected.